

Zihan Zhao

CONTACT INFORMATION

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INTRODUCTION

Ph.D. candidate in **Machine Learning** and **Particle Physics** at UC San Diego/CERN (LHC). I design **efficient transformer architectures**, with a focus on **sub-quadratic attention**, **long-context modeling**, and **latency-aware inference**, and I work on **self-supervised pretraining**, **efficient large-scale training**, and **LLM evaluation**. My work spans architecture design, distributed and mixed-precision training on GPU clusters, and reproducible open-source releases. I am seeking research roles building and advancing **frontier model** capabilities.

EDUCATION

University of California San Diego, San Diego, CA

Ph.D. in Physics, with emphasis on Machine Learning

Sept 2022 - Present

University of Southern California, Los Angeles, CA

B.S. in Physics; B.A. in Mathematics, Minor in Computer Programming

Aug 2018 - May 2022

- **GPA:** 3.97/4.0

WORK EXPERIENCE **Futurewei Technologies**

Machine Learning Research Intern, Intelligent Computing Lab

Jun 2025 - Sept 2025

Built **PULSE**, a privileged-knowledge distillation framework transferring representations from a rich teacher modality (EDA) to low-cost deployable sensors, used only at training time. Implemented per-modality masked-autoencoder encoders with shared/private embeddings, aligning student embeddings and intermediate states to a frozen teacher; ran ablations on mask ratio and modality. Trained on a multi-GPU V100 cluster with a reproducible release. Findings paper at **ML4H 2025**; proceedings paper accepted for **oral** at Sense of Space, **CVPR 2026**. **Outstanding Performance Intern**.

Contributed to **Deep-Reproducer**, a multi-agent LLM framework for automated paper reproduction; designed and built the self-evolving debugging loop (locate→patch→execute→test with fitness-based trajectory optimization). Co-authored paper at **NeurIPS 2025 Workshop on Deep Learning for Code (DL4C)**.

RESEARCH EXPERIENCE

PHAT-JeT: Patch Hierarchical Attention Transformer for Efficient Sequence Modeling

Graduate Student Researcher

Designed an attention architecture that reduces self-attention from $O(N^2)$ to $O(N)$ via patch-based local masked attention with hierarchical patch-token global communication, enabling longer contexts under fixed compute. Added geometric message passing (depthwise 2D convolution over the input grid) for local structure and a locality-preserving token ordering. Ran systematic ablations on ordering, message-passing variants, activations, and hierarchical-attention configs; benchmarked against full Transformer, Linformer, SAL-T, JEDI-Linear, and Point Transformer V3. Reached state-of-the-art accuracy at ~6.4k parameters with hardware-friendly operations targeting microsecond-latency inference. Co-authored manuscript **accepted by NeurIPS 2026**; reproducible implementation released.

SAL-T: Spatially Aware Linear Transformer for Efficient Inference

Graduate Student Researcher

Designed a linear-attention architecture using locality-preserving token ordering, linear partitioned key/value projections (LPP-MHA), and lightweight depthwise convolution over attention maps. Built the training and benchmarking stack; ran ablations over ordering, projection rank, and kernel size; profiled throughput, latency, and memory on consumer GPUs. Matched full-attention accuracy at linear complexity, with reduced FLOPs and inference latency and gains over Linformer. Co-authored manuscript submitted to **PRX Intelligence**; reproducible codebase released.

Conditional Capacity and Routing in Mixture-of-Experts Particle Transformers

Graduate Student Researcher

Studied sparse **Mixture-of-Experts (MoE)** versus dense Particle Transformers on 188-class JetClass-II, varying expert count, routing capacity factor, top-K, and auxiliary load-balancing loss. Showed that no-drop top-1 MoE improves accuracy by ~1 pp over the dense baseline at nearly unchanged forward FLOPs, while additional stored experts yield little further accuracy gain; top-2 routing improves prediction at ~51% more compute, and capacity-limited routing (up to 21% dropped assignments) accompanies degraded performance. Routing mutual-information analyses show expert assignments aligning with particle identity and kinematics, but not monotonically with classification performance. Manuscript submitted to **ML4PS @ NeurIPS 2026**.

Ill-Defined Math (IDM): Benchmarking LLM Reasoning Beyond Well-Defined Problems

Machine Learning Research Intern / Graduate Student Researcher

Led a multi-agent generation pipeline (Grok-4 generator, Gemini 2.5 Pro verifier, GPT-4o failure-filter) producing 1,300 ill-defined problems (300-problem expert-verified test set; 1,000-problem training set with distilled chain-of-thought). Built a three-stage LLM-as-judge framework (final-answer, issue recognition, issue fixing) reaching 1.00 / 0.94 F1 against human annotations. Benchmarked 27 frontier LLMs, surfacing a 54.4% abstention gap and systematic failure modes (guessing, hallucination, condition neglect). Co-authored manuscript **accepted by COLM 2026**.

Large-Scale Pretraining and Efficient Training for Foundation Models

Graduate Student Researcher

Built a pretraining/finetuning framework and studied how pretraining **dataset scale** affects downstream performance of transformer foundation models. Optimized throughput by reducing CPU-GPU synchronization, improving GPU matrix-multiplication kernel utilization, and using mixed-precision training. Presented at **ACAT 2024**; [arXiv:2408.09343](#).

J-JEPA: Self-Supervised Pretraining via Joint Embedding Predictive Architecture

Graduate Student Researcher

Adapted the Joint Embedding Predictive Architecture for set-structured inputs with redesigned masking, attention-based embeddings, and ordered positional encodings, yielding transferable representations under domain symmetries. Co-authored paper at **ML4PS @ NeurIPS 2024**; [arXiv:2412.05333](#).

ϕ^4 Lattice Field Theory: Regime-Dependent Failures of Gaussian Models

Mentor / Graduate Student Researcher

Mentored two undergraduates (A. Bhat, R. Ide) on a study of when Gaussian, basis-independent assumptions fail in 1D ϕ^4 theory. Showed failures arise from structured inter-mode dependencies rather than marginal non-Gaussianity, yielding a design criterion for nonlinear generative models. Co-authored paper accepted at **PAI 2026** (Stanford); [arXiv:2605.01145](#).

Information-Loss Audits of Learned Representations

Mentor / Graduate Student Researcher

Mentored high-school students Aiden Lee and Rishi Shah on auditing what information learned representations discard. Showed that “witness” tasks returned by task-discovery audits are largely

determined by the task-generator class rather than the representation: on 130,000 simulated LHC jets, class-mean gaps span 0.622–0.923 within a single encoder, and returned tasks are $2.49\text{--}3.25\times$ as similar within a generator class as across classes. Proposed a witness-set reporting protocol that recovers the planted factor on 3/4 encoders, where the single top-scoring witness misses it. Manuscript submitted to **ML4PS @ NeurIPS 2026**.

Higgs Boson Measurement ($H \rightarrow b\bar{b}$), LHC Run 3

Graduate Student Researcher

Applied ML to a CMS physics analysis: jet-tagging efficiency, decision boundaries between production modes, and neural-network background estimation, on large collision datasets (ROOT, Pandas, NumPy).

TALKS AND PRESENTATIONS

22nd International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT 2024)

Gave a contributed talk on **Self-Supervised Learning for Jet Tagging** at Stony Brook University, NY, USA.

ML4Jets Workshop 2024

Gave a contributed talk on **Large-Scale Pretraining and Finetuning for Efficient Jet Classification** at LPNHE, Paris, France

NeurIPS Workshop on Machine Learning for Physical Sciences

Presented on **Learning Symmetry-Independent Jet Representation via Jet-Based Joint Embedding Predictive Architecture** at Vancouver, BC, Canada.

Larger than Larger: Large AI Models at the Frontiers of Experimental High-Energy Physics. Episode I: Large Jet Models

Gave a contributed talk on **Large-Scale Pretraining and Finetuning for Efficient Jet Classification and Learning Symmetry-Independent Jet Representation via Jet-Based Joint Embedding Predictive Architecture** at Peking University, Beijing, China

Fast Machine Learning for Science Conference 2026 (FastML 2026)

Gave a contributed talk on **Efficient Attention Architectures for Real-Time Jet Tagging at the LHC: SAL-T and PHAT-JeT** at San Diego, CA, USA. **Received the best talk award**

PUBLICATIONS

H. Chen*, Y. Lin*, **Z. Zhao***, P. Chen*, et al. "Ill-Defined Math: Benchmarking LLM Reasoning Beyond Well-Defined Problems." **COLM 2026**

P. Chen, N. Yan, **Z. Zhao**, Y. Lin, H. Chen, Y. Hu, Q. Bai, X. Li, M. S. Mortazavi. (2025). "Deep-Reproducer: From Paper Understanding to Code Generation." Workshop on Deep Learning for Code in the Agentic Era (DL4C) @ NeurIPS 2025. **OpenReview**

Z. Zhao*, A. Wang*, A. Xia*, A. Gandrakota, J. Ngadiuba, R. Cavanaugh, J. Duarte. "PHAT-JeT: Patch Hierarchical Attention Transformer for Efficient Particle Jet Tagging.". **NeurIPS 2026** [arXiv:2605.21789](https://arxiv.org/abs/2605.21789)

A. Bhat, R. Ide, **Z. Zhao**. (2026). "When Independent Gaussian Models Break Down: Characterizing Regime-Dependent Modeling Failures in ϕ^4 Theory." Physics $\cap$ AI Symposium (PAI 2026), Stanford University. [arXiv:2605.01145](https://arxiv.org/abs/2605.01145)

T. Legge, A. Wang, J. Ortiz, V. Limouzi, **Z. Zhao**, A. Gandrakota, E. E. Khoda, J. Ngadiuba, J. Duarte, R. Cavanaugh. (2025). "Why Is Attention Sparse In Particle Transformer?" [arXiv:2512.00210](https://arxiv.org/abs/2512.00210),

Z. Zhao, K. Pendiyala, M. Mortazavi, N. Yan. (2026). “PULSE: Privileged Knowledge Transfer from Rich to Deployable Sensors for Embodied Multi-Sensory Learning” Sense of Space Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026. [arXiv:2510.24058](#)

Z. Zhao*, A. Wang*, S. Katel, V. G. Sahu, E. E. Khoda, A. Gandrakota, J. Ngadiuba, R. Cavanaugh, J. Duarte. (2025). “Spatially Aware Linear Transformer (SAL-T) for Particle Jet Tagging.” [arXiv:2510.23641](#)

Z. Zhao*, S. Katel*, H. Li*, J. Duarte (2024). Learning Symmetry-Independent Jet Representations via Jet-Based Joint Embedding Predictive Architecture. Machine Learning and the Physical Sciences Workshop @ NeurIPS 2024 [arXiv:2412.05333](#)

Z. Zhao et al., “Large-Scale Pretraining and Finetuning for Efficient Jet Classification in Particle Physics,” in *22nd International Workshop on Advanced Computing and Analysis Techniques in Physics Research: Foundation Models for Physics - Nexus of Computation and Physics through Embracing the Era of Foundation Models*, 8, 2024. [arXiv:2408.09343](#).

AWARDS

Dean’s List all semesters, USC	2018 - 2022
Member of Alpha Lambda Delta National Honor Society, USC	2018
Dornsife Continuing Student Scholarship, USC	2019, 2020, 2021
Dornsife Student Opportunities for Academic Research Scholarship, USC	2021
Recipient of Dornsife Summer Undergraduate Research Fund, USC	2021
GPSA Travel Grant Award, UC San Diego	2024
Chairs Challenge Award, UC San Diego	2024

TEACHING AND MENTORING

Teaching Assistant, UC San Diego: PHYS 2C (Waves, Thermodynamics, Optics), PHYS 4C (E&M), PHYS 4E (Intro Quantum). Led weekly discussion sections.

Student research mentoring, UC San Diego: scoped and supervised ML research projects.

- **Subash Katel** — J-JEPA → Ph.D. at **Princeton** (Quantum Computing & Computer Architecture).
- **Molan Li** — n-ParT (Normalized Particle Transformer) → Ph.D. at **UIUC** (Quantum Information).
- **Anish Bhat, Ryo Ide** — ϕ^4 lattice study → co-first-authored paper at **PAI 2026** (Stanford); [arXiv:2605.01145](#).
- **Aiden Lee, Rishi Shah** (high school) — Information-loss audits of learned representations → paper submitted to **ML4PS @ NeurIPS 2026**.
- **Kaushik Pendiyala** — PULSE (privileged knowledge transfer) and MoE Particle Transformers → co-authored PULSE, **oral** at Sense of Space, **CVPR 2026**; MoE paper submitted to **ML4PS @ NeurIPS 2026**.

COMMUNITY SERVICE

- **Conference Reviewer**: ICLR 2027
- **Symposium Reviewer**: Physics $\cap$ AI Symposium (PAI 2026), Stanford University
- **Workshop Reviewer**: ML4PS @ NeurIPS 2026, NeurIPS 2026 Interp4Discovery, NeurIPS 2026 MATH-AI

OUTREACH

Volunteered at Duarte Lab / UCSD Physics booths, **Southeast San Diego Art and Science Expo** and **Barrio Logan Science and Art Expo**, communicating physics concepts to K–12 audiences through hands-on demos.

SKILLS

- Modeling & training: PyTorch; multi-GPU distributed training, mixed precision, CUDA-aware optimization
- Languages: Python, C++, Java, MATLAB, SQL
- Tools & libraries: Docker, Kubernetes, Slurm; NumPy, Pandas, H5py, Matplotlib